

# Chapter 1, Section 7

James Lee

September 11, 2026

1. (a) Find the degree of the  $d$ -uple embedding of  $\mathbb{P}^n$  in  $\mathbb{P}^N$ .
- (b) Find the degree of the Segre embedding of  $\mathbb{P}^r \times \mathbb{P}^s$  in  $\mathbb{P}^N$ .

*Proof.*

- (a) The homogenous coordinate ring of the  $d$ -uple embedding is the  $k[y_0, \dots, y_n]$  where each  $y_i$  has degree  $d$  as a graded  $k[x_0, \dots, x_n]$ -module. The Hilbert polynomial of the  $d$ -uple embedding is

$$P(z) = \binom{n + dz}{n} = \frac{(dz + n)(dz + n - 1) \cdots (dz)}{n!} = \frac{d^n}{n!} z^n + \cdots,$$

Hence, the degree is  $d^n$ .

- (b) We have seen in (Ex. 2.14) that the homogenous coordinate ring of the Segre embedding is isomorphic to  $k[\{x_i y_j\}]$ , which is a subring of  $k[x_0, \dots, x_r, y_0, \dots, y_s]$ . The grading is given by  $\bigoplus_{k=0}^{\infty} M_k$  where  $M_k$  is the set of polynomials of degree  $2k$  in  $k[\{x_i y_j\}]$ . Since each monomial is made up of half  $x_i$ 's and half  $y_j$ 's, the Hilbert polynomial is

$$P(z) = \binom{r + z}{r} + \binom{s + z}{s} = \frac{1}{r!s!} z^{r+s} + \cdots,$$

hence the degree is  $\binom{r+s}{r}$ .

□

3. *Dual Curve.* Let  $Y \subseteq \mathbb{P}^2$  be a curve. We regard the set of lines in  $\mathbb{P}^2$  as another projective space,  $(\mathbb{P}^2)^*$ , by taking  $(a_0, a_1, a_2)$  as homogenous coordinates of the line  $L : a_0 x_0 + a_1 x_1 + a_2 x_2 = 0$ . For each nonsingular point  $P \in Y$ , show that there is a unique line  $T_P(Y)$  whose intersection multiplicity with  $Y$  at  $P$  is  $> 1$ . This is the *tangent line* to  $Y$  at  $P$ . Show that the mapping  $P \mapsto T_P(Y)$  defines a *morphism* of  $\text{Reg } Y$  (the set of nonsingular points of  $Y$ ) into  $(\mathbb{P}^2)^*$ . The closure of the image of this morphism is called the dual curve  $Y^* \subseteq (\mathbb{P}^2)^*$  of  $Y$ .

*Proof.* Let  $x_0, x_1, x_2$  be homogenous coordinates for  $\mathbb{P}^2$ , and let  $f$  be an irreducible homogenous polynomial in  $S = k[x_0, x_1, x_2]$  that defines  $Y$  and let  $P = (c_0, c_1, c_2)$  be a nonsingular point on  $Y$ . Then  $\mathbb{P}^2$  can be covered by  $U_i$  where  $U_i \cong \mathbb{A}^2$  and is defined by  $x_i \neq 0$ . Since every  $P$  is contained in some  $U_i$ , we find an affine line in  $U_i$  that is tangent to the affine variety  $Y_i := Y \cap U_i$ . We claim that the projective closure of this line is the desired unique line  $T_P(Y)$ .

Without loss of generality, assume  $c_0 = 1$  so that  $P \in U_0$ . Identifying  $U_0$  with  $\mathbb{A}_k^2$  with coordinates  $y_1, y_2$ , we have  $P_0 = (c_1, c_2) \in \mathbb{A}_k^2$  identified with  $P$ , and  $Y_0$  is defined by the vanishing of  $g(y_1, y_2) = f(1, y_1, y_2)$ . Assume  $c_1 = c_2 = 0$  for simplicity. Let  $\mathfrak{m}_{P_0}$  be the maximal ideal in  $S(U_0) \simeq k[y_1, y_2]$  corresponding to the point  $P_0$ . Since  $Y$  is nonsingular at  $P$ ,  $Y_0$  is nonsingular at  $P_0$ , which means  $a_i = (\partial g / \partial y_i)(P_0)$  for  $i = 1, 2$  are not both zero. In particular, we have  $g \notin \mathfrak{m}_{P_0}^2$ , so we can write

$$g = a_1 y_1 + a_2 y_2 + (\text{higher degree terms}),$$

thus any linear polynomial in  $\mathfrak{m}_P^2 + (g)$  must be a constant multiple of  $t = a_1 y_1 + a_2 y_2 = 0$ . We define  $T_P(Y)$  to be the projective closure of  $t$ . We show it is the desired unique line by showing if a line  $H : h(x_0, x_1, x_2) = b_0 x_0 + b_1 x_1 + b_2 x_2 = 0$  has intersection multiplicity with  $Y$  at  $P > 1$ , then  $H \cap U_0 = T$ , which implies

$T_P(Y) = H$  since  $H \cap U_0 = T$  is dense in both  $H$  and  $T$ . Setting  $S(Y) = S/(f)$  and  $\mathfrak{m}_P$  as the homogenous prime ideal in  $S$  corresponding to the point  $P \in \mathbb{P}^2$ , we have  $S/(f, h) = S(Y)/(h)$ , thus

$$\begin{aligned} i(Y, H; P) > 1 &\iff (S(Y)/(h))_{\mathfrak{m}_P} \text{ has length } k > 1 \text{ as an } S(Y)_{\mathfrak{m}_P}\text{-module} \\ &\iff \mathfrak{m}_P/(h) \text{ has length } k \text{ as an } S(Y)\text{-module} \\ &\iff (h) = \mathfrak{m}_P^k \text{ as an extended ideal in } S(Y)_{\mathfrak{m}_P} \end{aligned}$$

since  $S(Y)_{\mathfrak{m}_P}$  is a discrete valuation ring by non-singularity of  $Y$  at  $P$ . In terms of affine coordinates, let  $\ell = b_0 + b_1y_1 + b_2y_2$  so that  $H \cap U_0$  can be identified with the affine variety defined by the vanishing of  $\ell$ , where  $b_0 = 0$  necessarily since  $P_0 = (0, 0)$  is a root of  $\ell$ . Then, by the statements above, we have  $\ell \in \mathfrak{m}_{P_0}^k + (g) \subseteq \mathfrak{m}_{P_0}^2 + (g)$ , which is possible if and only if  $\ell = \lambda t$  for some  $\lambda \neq 0$ , hence  $H \cap U_0 = T$ .  $\square$

4. Given a curve  $Y$  of degree  $d$  in  $\mathbb{P}^2$ , show that there is a nonempty open subset  $U$  of  $(\mathbb{P}^2)^*$  in its Zariski topology such that for each  $L \in U$ ,  $L$  meets  $Y$  in exactly  $d$  points. This result shows that we could have defined the degree of  $Y$  to be the number  $d$  such that almost all lines in  $\mathbb{P}^2$  meet  $Y$  in  $d$  points, where "almost all" refers to a nonempty open set of the set of lines, when this set is identified with the dual projective space  $(\mathbb{P}^2)^*$ .

*Proof.* Following the hint, we show that the set of lines in  $(\mathbb{P}^2)^*$  which are either tangent to  $Y$  or pass through a singular point of  $Y$  is contained in a properly closed subset. By the previous exercise, a line can meet  $Y$  at exactly  $d$  points if and only if it has intersection multiplicity equal to 1 at every point of intersection, thus the set of lines which are either tangent to  $Y$  or pass through a singular point of  $Y$  is contained in the dual curve  $Y^*$  in  $(\mathbb{P}^2)^*$ , hence the set of lines that meet  $Y$  at exactly  $d$  points is contained in the open subset  $(\mathbb{P}^2)^* - Y^*$ .  $\square$